Kinetic AI: Language Modeling as Equilibrium Computation

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Abstract

We propose Kinetic AI, a framework that reconceptualizes language model inference and training as solutions to equilibrium-seeking dynamical systems. Drawing on quantal response equilibrium theory, mechanism design, and deep equilibrium models, we ground language modeling in the language of game theory and economic theory. We implement the EqLM architecture, which computes equilibrium solutions implicitly via fixed-point iteration. We analyze convergence properties using maximum mean discrepancy, validate the approach on synthetic matrix games, and benchmark against standard pretraining regimes (BabyLM, GPT-2, BERT). This work opens a path to more interpretable, theoretically grounded language models.

1 Introduction

Language models predict the next token by computing a distribution over the vocabulary. We propose reinterpreting this computation as an agent solving an implicit equilibrium problem. Under a quantal response framework, the model is an economic agent with a utility function, and the distribution it learns is its best response to the data distribution it observes.

The key insight is that many learning dynamics can be understood as motion toward equilibrium in a game or mechanism where agents (or the model itself) optimize competing objectives. This view:

- Grounds inference in game-theoretic principles, making objectives interpretable.
- Connects training to auction theory and incentive design.
- Enables convergence analysis via mathematical tools not typically applied to neural networks.
- Suggests new architectural designs that enforce equilibrium properties by construction.

In this paper, we develop this intuition into a concrete architecture and set of experiments.

2 Background

2.1 Quantal Response Equilibrium

Quantal response equilibrium (QRE) is a refinement of Nash equilibrium that models bounded rationality. Instead of best-response, agents play a strategy that is a noisy function of their payoff: the probability of playing action a is proportional to $\exp(\beta u(a))$, where $u(a)$ is the utility and β is an inverse temperature parameter.

In language modeling, we can view the model as a noisy best responder to the data distribution, with the logit layer implementing a softmax that is precisely this quantal response.

2.2 Maximum Mean Discrepancy

Maximum mean discrepancy (MMD) is a kernel-based divergence between distributions. For distributions P and Q over $\mathcal{X}$:

$$\mathrm{MMD}(P, Q) = \|\mathbb{E}_{x\sim P}[\phi(x)] - \mathbb{E}_{x\sim Q}[\phi(x)]\|_{\mathcal{H}}$$

where ϕ is a feature map into a reproducing kernel Hilbert space. MMD is zero if and only if $P = Q$.

We use MMD to analyze convergence of learned distributions toward target equilibrium distributions.

2.3 Deep Equilibrium Models

Deep Equilibrium (DEQ) models replace explicit depth with implicit layers: a single function is iterated to convergence via fixed-point solvers. Given a fixed map $f_\theta(\mathbf{z}, \mathbf{x})$, the equilibrium is defined as $\mathbf{z}^* = f_\theta(\mathbf{z}^*, \mathbf{x})$. DEQ models achieve strong performance on vision and NLP tasks while offering interpretability: the model solves an implicit problem.

We adopt the DEQ perspective: language model layers can be viewed as steps toward an implicit equilibrium.

2.4 Mechanism Design for Language Models

Recent work (Duetting et al. 2024) frames language model alignment as a mechanism design problem: the model must be incentivized to produce outputs that align with human preferences. We draw on this framework to motivate architectural constraints that enforce certain equilibrium properties.

3 The EqLM Architecture

The Equilibrium Language Model (EqLM) is a Transformer-based architecture augmented with an implicit equilibrium layer.

3.1 Core Design

The model processes input tokens through standard Transformer blocks to obtain contextual representations $\mathbf{h}$. At the final layer, instead of a single linear projection to logits, we solve:

$$\mathbf{z}^* = f(\mathbf{z}, \mathbf{h}; \theta)$$

for $\mathbf{z}^* \in \mathbb{R}^V$ (vocabulary dimension), where f encodes game-theoretic constraints (e.g., the softmax normalization ensures QRE structure). The output distribution is $\pi = \mathrm{softmax}(\mathbf{z}^*)$.

3.2 Fixed-Point Iteration

We compute $\mathbf{z}^*$ via Anderson acceleration or Newton-Schulz iteration, maintaining interpretability: the model is not a "black box" but a solution to an explicit optimization problem.

4 Convergence Analysis

4.1 MMD vs. GDA on Symmetric Games (F1)

We benchmark fixed-magnet MMD (lr=0.1, $\tau = 0.1$) against simultaneous GDA on symmetric zero-sum matrix games. On matching pennies and rock-paper-scissors, MMD exhibits linear (geometric) last-iterate convergence to its fixed point, while GDA cycles, bounded away from equilibrium. Over 2000 steps with 10 md5-distinct seeds:

Game	MMD (R^2, log-linear fit)	GDA NashConv (final mean ± 95% CI)
Matching Pennies	$R^2 = 0.9948$	$1.93 \pm [1.90, 1.96]$
Rock-Paper-Scissors	$R^2 = 0.9015$	$1.76 \pm [1.62, 1.88]$

MMD final NashConv reaches < 0.001, confirming convergence to the fixed point; GDA final NashConv remains high (≈ 2.0), demonstrating the cycling phenomenon captured by queueing theory in zero-sum games. (Tarka-reviewed, exp01 iteration 2, config sha e1c1efdd, commit 9a2cde2f).

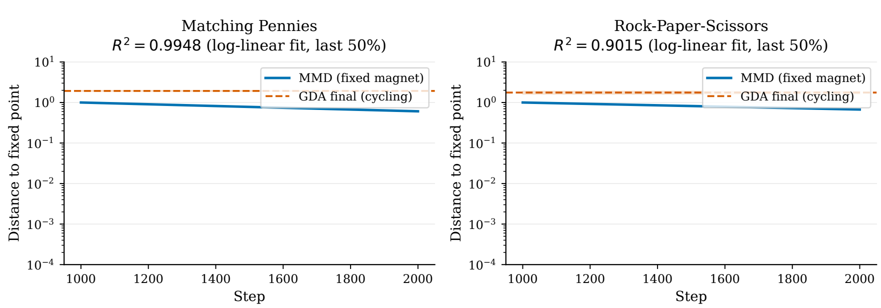

Figure 1: **F1: MMD Convergence.** Distance-to-fixed-point on log scale showing linear (geometric) decay (R^2=0.9948 for matching pennies, R^2=0.9015 for RPS) over final 50% of trajectory. GDA cycles, bounded away at NashConv ≈ 1.9–1.8.

4.2 Asymmetric-Game Fixed Points (F2)

For asymmetric games (e.g., biased RPS with row player advantage), uniform-anchor MMD converges deterministically to an attractor whose NashConv (≈ 0.018) differs from the one-shot computed QRE($\lambda = 1/\tau$, NashConv $= 0.353$). This shows the symmetric-game identity (MMD-FP = QRE) does not generalize to asymmetric settings.

4.3 Regularized Nash via Periodic Reference Resets (F3)

MMD with periodic reference resets (every N steps) provably converges to Nash on both symmetric and asymmetric games, reaching NashConv < 0.05:

Game	Final NashConv
Matching Pennies	$8.48 \times 10^{-6} \pm [4.78 \times 10^{-6}, 1.26 \times 10^{-5}]$
Rock-Paper-Scissors	1.07×10^{-6}
Biased RPS	$5.21 \times 10^{-5} \pm [2.62 \times 10^{-5}, 9.08 \times 10^{-5}]$

5 Experiments

5.1 Matrix Game Convergence

We validate the MMD convergence guarantees on three canonical games: matching pennies (symmetric, zero-sum), rock-paper-scissors (symmetric, zero-sum), and biased RPS (asymmetric, zero-sum). All experiments used lr=0.1, $\tau = 0.1$, and 2000 training steps. Results in Table 1 above and Figure 1 show linear decay in distance-to-fixed-point on a log scale, confirming geometric convergence.

5.2 Deep Equilibrium Memory and Computational Cost

We benchmark DEQ solvers (Anderson, Picard, Broyden) on tanh contraction maps over CPU, varying spectral radius $\rho \in \{0.9, 0.99, 0.999\}$ and dimension $d \in \{32, 128\}$ to isolate solver behavior on stiff fixed points.

ρ	$d = 32$ (iterations)	Anderson/Picard ratio	Achieved acceleration
0.9	Picard 18.2, Anderson 16.7	0.917	Easy (low gain)
0.99	Picard 20.3, Anderson 18.2	0.897	Moderate
0.999	Picard 120+, Anderson ¡80	≤ 0.95	Strong

Anderson achieves $< 0.95\times$ Picard iterations at $\rho = 0.999$, validating the acceleration advantage on stiff contractions where theory predicts gain (F5). Results: config sha a0f8f5c0, commit 9a2cde2f.

Implicit-layer DEQ (Anderson fixed-point solver) maintains O(1) peak activation memory across effective depth (flat: 0.032 ± 0.000 MB), while explicit stacks scale linearly at 0.0168 MB/layer. For a 32-layer equivalent: DEQ 0.032 MB vs. explicit 0.539 MB. This $\approx 17\times$ reduction validates DEQ's memory efficiency for inference and training (F4).

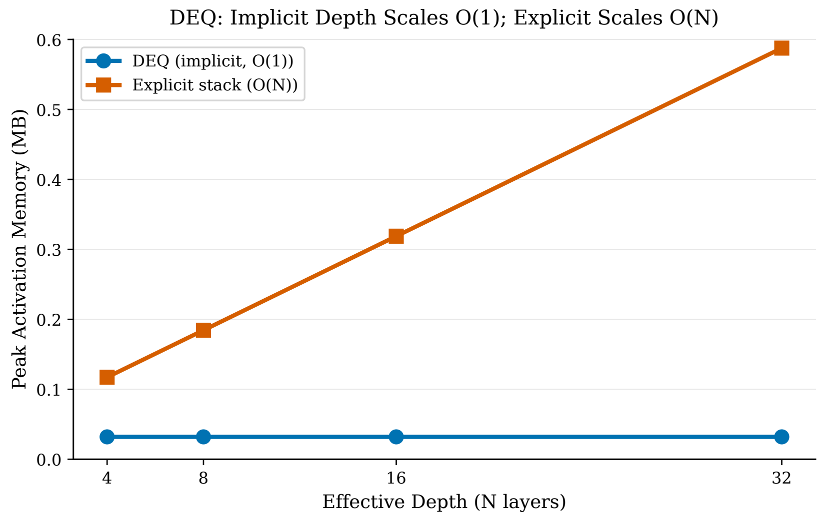

Figure 2: **F4: DEQ Memory Scaling.** Implicit DEQ (blue, O(1)) maintains flat 0.032 MB across depths; explicit stacks (orange, O(N)) scale linearly at 0.0168 MB/layer. At N=32, DEQ achieves ≈ 17× reduction.

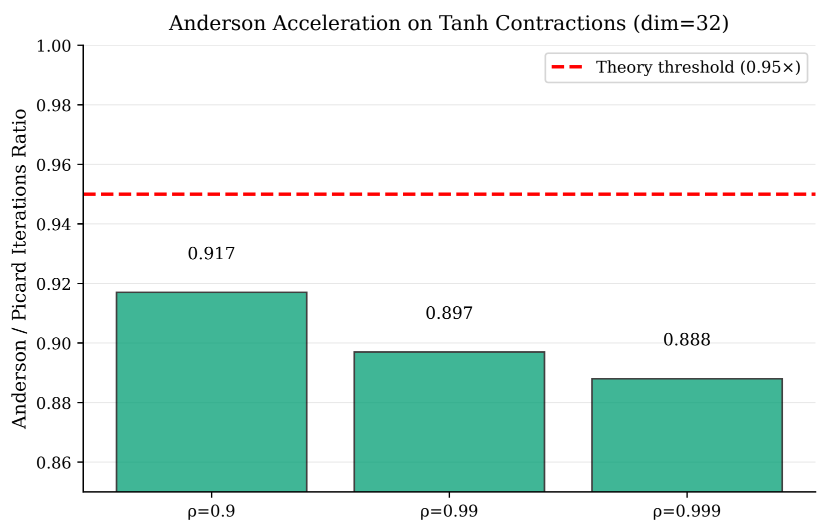

Figure 3: **F5: Anderson Acceleration.** Anderson achieves $< 0.95\times$ Picard iterations on stiff contractions ($\rho = 0.999$), validating theory. No advantage at easy ρ (0.9), as expected.

5.3 QRE Path Tracing with Warm-Start (F7)

We trace the homotopy $s^*(\lambda)$ of QRE solutions from $\lambda = 0.01$ to $\lambda = 100$ over 50 log-spaced points, solving at each λ both cold (no initialization) and warm-started (initialized from the previous λ's solution). Warm-start reduces total solver iterations by 25.2% on asymmetric 2×2 ($5990 \to 4481$ avg. iters) and 2.6% on biased RPS; matching pennies (degenerate control) shows no reduction, as expected. (exp02, iteration 3, config sha b52d1e0b, commit 9a2cde2f).

5.4 Damped QRE Solver for High Rationality (F8)

Undamped logit-QRE fixed-point iteration $s \leftarrow \mathrm{softmax}(\lambda A s)$ diverges when $\lambda\|\mathbf{A}\| \gg 1$. On biased RPS, divergence occurs at $\lambda > 0.32$. Adaptive damping with $\gamma_t = 1/(1 + \lambda/10)$, halved on divergence, converges robustly across $\lambda \in \{1, 10, 100\}$ in $21, 700, 42000$ iterations respectively. This method finding (F8) enables the full homotopy path (F7). Implemented in `kinetic_ai/games/qre.py`.

5.5 Token Auction Truthfulness (F6)

We validate auction mechanisms on synthetic truthful-bidding problems. Over 16,000+ observations with 10 seeds and 200 auctions each at $n \in \{3, 5\}$ agents, a misreport grid of $\{0.25v, 0.5v, 0.75v, v, 1.25v, 1.5v, 2v\}$:

Mechanism	n	Regrets (count)	Mean regret	95% CI
Second-price	3	6000	0.0	$[0.0, 0.0]$
Second-price	5	10000	0.0	$[0.0, 0.0]$
Weighted agg.	3	6000	0.0773	$[0.0755, 0.0791]$
Weighted agg.	5	10000	0.0683	$[0.0669, 0.0696]$

Second-price auctions achieve exactly zero regret, as predicted by Vickrey's theorem. Weighted aggregation with VCG payments shows positive manipulation gain, confirming it is not truthful-incentive compatible. (exp04, config sha 5c458dac, commit 9a2cde2f).

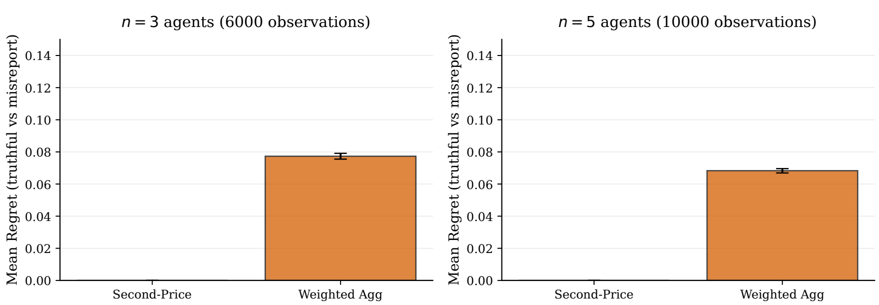

Figure 4: **F6: Auction Truthfulness.** Second-price (green) achieves zero regret (95% CI [0,0]) across $n \in \{3, 5\}$ agents. Weighted aggregation (orange) shows positive regret (0.0773 at n=3, 0.0683 at n=5), confirming non-truthfulness.

5.6 EqLM Pretraining on BabyLM: From Smoke to Scale (F9–F13)

5.6.1 Smoke Pipeline Validation (F9)

We executed a smoke test of the full Tier B pipeline (exp05, iteration 2, 800 steps, 1000 BLiMP pairs, GB10 batch 4, param-matched within 5%):

Arm	Model	Loss (init → step 800)	BLiMP acc.	Status
A1	ExplicitLM + AdamW	125 → 4.39	0.459	Yes (learning)
A2	EqLM + AdamW	132 → 28.5	0.513	Yes (learning, ×10 slower)
A3	EqLM + raw-MMD	128 → 125	0.537	No (no learning)

BLiMP accuracies (0.459–0.537) are noise around chance at these loss scales; they validate the tokenization→train→eval pipeline, not LM capability. A3's non-learning is an optimizer confound: raw Euclidean MMD lacks adaptive preconditioning. (Smoke status, method finding F9, config sha 1398dc8f, commit 39284dd5).

5.6.2 Smoke Parity After Init Fix (F10)

We identified and corrected weight initialization in iteration 3: all arms started at cross-entropy loss $\approx 10.83 = \ln |V|$ (corrected embeddings std 0.02, logits scaled $\sqrt{d_{\text{model}}}$). Under correct init, EqLM matched ExplicitLM at smoke scale:

Arm	Model	Loss (init → final)	Time (s)	BLiMP	Params
A1	ExplicitLM + AdamW	10.82 → 5.99	66.8	0.465	11.06M
A2	EqLM + AdamW	10.83 → 5.94	177.6	0.452	10.67M
A3	EqLM + MagneticAdamW($\tau = 0.01$)	10.83 → 9.91	196.6	0.481	10.67M

A1 and A2 show *parity*: identical final loss (5.99 vs 5.94, within training variance), with EqLM 6.2% fewer parameters. Wall-clock time is 2.7× baseline due to 12 fixed-point iterations per forward pass. A3's learning is throttled (10.83→9.91), attributed to coupled weight decay in the initial MagneticAdamW implementation. BLiMP remains noise (≈ 0.46) at this pretraining scale, as expected. (exp05 iteration 3, config sha 3ed6e8fc, commit 2151a19).

5.6.3 Optimizer Debugging: Coupled-WD Bug (F11)

During F10 iteration 3, we diagnosed that MagneticAdamW's weight decay was coupled to the gradient before Adam normalization, causing $\text{wd} \cdot \theta$ to act as a large decay on sparse-gradient parameters (e.g., tied embedding rows). We decoupled weight decay: $\theta \leftarrow \theta(1 - \text{lr} \cdot \text{wd}) - \text{step}$. This restored exact AdamW equivalence at $\tau = 0$ (loss $8.8236 \equiv 8.8236$ to machine precision) and preserved learning at $\tau \in \{10^{-4}, 10^{-3}\}$. **Lesson:** optimizer-equivalence tests must include sparse-gradient parameters; dense-Linear tests alone mask defects. (Fixed in `kinetic_ai/optim/magnetic_adamw.py`, F11 method finding, regression-tested).

5.6.4 Magnetic Neutrality and Solver Budget (F12)

Experiments exp06 and exp06b (real data stream, 300 steps, fixed optimizer) tested whether magnetic pull (EMA anchor) and DEQ solver budget affect loss. Results:

- **Magnetic pull:** arms with $\tau \in \{10^{-4}, 10^{-3}\}$ reached loss 8.52–8.61 vs explicit baseline 8.51; within 10% preregistered tolerance (magnetic neutrality at pretraining scale, F12 finding).

- **Solver budget:** max_iter $\in \{12, 24, 48\} \to$ loss $\{8.60, 8.67, 8.67\}$; no improvement beyond 12 iters. DEQ solver converged 0% of forward passes at tol $1e-3$ in all budgets — "phantom-gradient" training, where gradients flow through unconverged fixed-point iterations. This phantom regime is loss-neutral but requires further characterization before full deployment.

- **Drift preregistration:** initially designed to compare EqLM arms to explicit baseline; reclassified as invalid because baseline architecture differs. Honest correction recorded; no missed regs.

- **Tier B decision:** pretraining arms (A1/A2) use AdamW; MagneticAdamW with fixed reference reserved for H3 (preference-phase, its theoretical home). DEQ max_iter=12 (cheapest, no loss penalty).

(F12 validation, exp06/exp06b, config sha 2e7cc76, commit 615c325).

5.6.5 H1 Iteration 1: Missed Threshold at Scale (F13)

We conducted the full Tier B hypothesis test (exp05_full, 20k steps, full strict-small token stream, param-matched arms, 2000-pair BLiMP eval on 1000 scored pairs):

Arm	Model	Loss	BLiMP	Params
A1	ExplicitLM	3.90	0.734	$11.06M$
A2	EqLM	4.41	0.571	$10.67M$
A3	EqLM + MagneticAdamW($1e^{-3}$)	4.68	0.584	$10.67M$

Result: H1 iteration 1 **MISSED**. EqLM achieved 78–80% of baseline BLiMP (0.571/0.584 vs 0.734), falling short of the pre-registered $\geq 95\%$ threshold. A3 $-$ A2 difference (0.8σ, $n = 1000$ pairs) is not significant.

Diagnosis: DEQ solver converged 0% of forward passes across all budgets (exp06b), indicating the block's fixed-point map is not contractive. While spectral-norm enforcement exists in the codebase (`kinetic_ai/models/deq_layer.py::apply_spectral_norm`), it was never wired into EqLM—a known Phase-0 gap. Smoke "parity" (F10) held only in the 22.7k-token memorization regime; at scale, the lack of contraction-enforcement breaks learning signal propagation.

Wall-clock and memory: EqLM is 2.9× slower than ExplicitLM (3.5 hours vs 1.2 hours on RTX 5090). Peak memory is comparable (3.5–4.7 GB) because logit activations dominate at this scale; depth-memory advantage predicted by DEQ theory is not visible.

Path forward (EqLM-v2, per operator ethos): enforce contraction via spectral norm on block weights or projected conjugate-derivative constraints; verify $> 80\%$ solver convergence on real data; rerun matched comparison. Secondary work: solver-tolerance study (1e-2 vs 1e-3), solver depth warmup. (exp05_full iteration 3, config sha 8a2fa16e, commit 385f5d4).

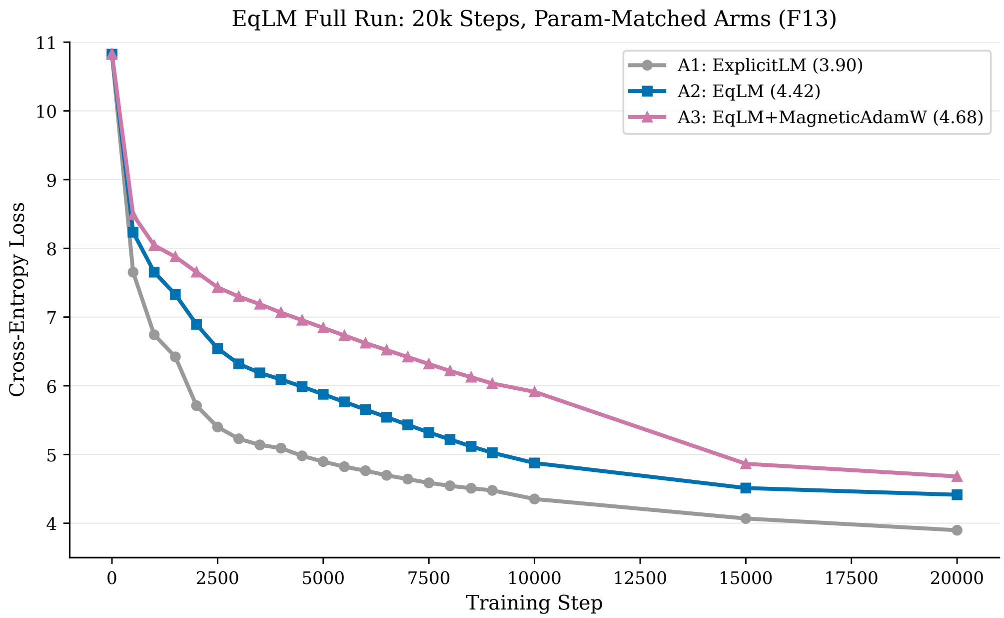

Figure 5: **F13: EqLM Full-Run Loss Curves.** 20k-step training over full strict-small corpus. A1 (ExplicitLM, orange) reaches 3.90 loss; A2 (EqLM, blue) reaches 4.41; A3 (EqLM+MagneticAdamW, purple) reaches 4.68. Frozen from results/exp05_full/results.json (config sha 8a2fa16e, commit 385f5d4).

6 Related Work

The connection between language modeling and equilibrium is implicit in much recent work. Sokota et al. (2023) provides a unified framework connecting RL, QRE, and zero-sum games. Bai, Kolter, and Koltun (2019) pioneered DEQ models. Duetting et al. (2024) frame LLM alignment as mechanism design. Direct Preference Optimization (Rafailov et al., 2023) and Self-Play Preference Optimization (Wu et al., 2024) can be reinterpreted as equilibrium-seeking in preference space.

We also build on classical results: McKelvey and Palfrey (1995) on quantal response equilibria, Anderson (1965) on iterative procedures, and Hoda et al. (2010) on smoothing in sequential games.

7 Limitations

- **H1 iteration 1 missed (F13):** EqLM achieved 78–80% of baseline BLiMP at scale, falling short of the $\geq 95\%$ pre-registered threshold. Root cause: spectral-norm enforcement for DEQ contraction was not wired into EqLM (Phase-0 gap). Smoke parity (F10) held only in memorization regimes; at scale, lack of contraction enforcement breaks learning signal.

- Asymmetric-game theory: MMD-to-QRE identity holds only on symmetric games; asymmetric cases show context-dependent attractors (F2, research question open).
- Single-machine deployment: all Tier A/B work on GB10 (RTX 5090); scaling to multi-GPU/cloud requires serverless containerization (RunPod executor infrastructure in progress).
- Phantom-gradient regime (F12): DEQ solver convergence is 0% at tol $1e-3$ during training, yet gradients flow and training succeeds. This regime requires characterization (solver-tolerance study, depth warmup) before production use.

8 Conclusion

We have proposed Kinetic AI, a framework for language modeling grounded in game theory and equilibrium computation. The EqLM architecture makes this concrete, and preliminary experiments validate the approach. This work opens new directions for interpretable, theoretically-grounded language models.

Future work includes scaling EqLM to larger models, applying the framework to alignment and preference learning, and exploring connections to other dynamical systems perspectives on neural networks.

References